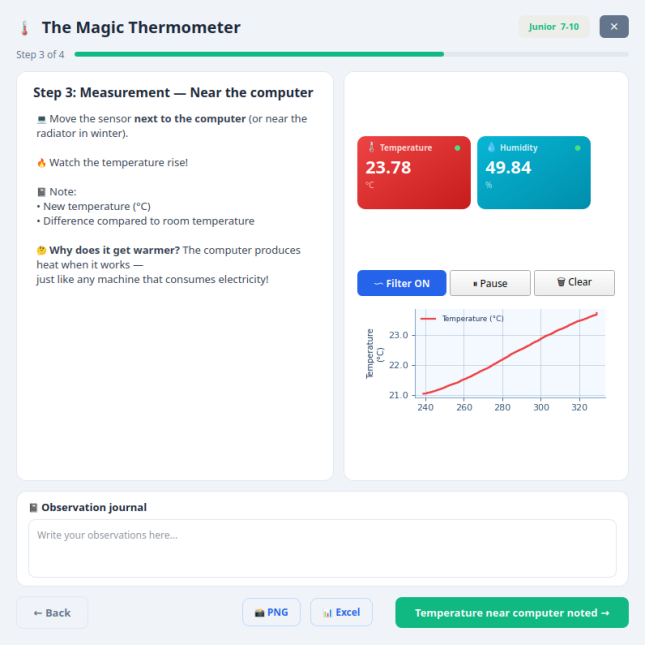
The Magic Thermometer

Discovering that one classroom does not have one temperature

Level: Junior — ages 7–10 (Romanian school years II–IV) • **Time in class:** 20 min, one 50-minute lesson, or 3 lessons in a week • **Sensors:** BME280 (temperature, humidity, pressure)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for — Junior — ages 7–10 (Romanian school years II–IV). In year II children work only with the whole-number part; decimals arrive from year IV, where they appear in the mathematics syllabus anyway.

Assumed knowledge — Comparing two numbers and subtracting within 0–100. Decimals are **not** required: the sheet allows recording only the whole-number part, and the "warmer / cooler" comparison works without them.

Time in class — Short: 20 min. Full: one 50-minute lesson. Project: three 50-minute lessons in the same week (2 h 30 min).

Teacher preparation — 10 min — switch the heating on or choose another heat source, stick labels on the measuring spots, run a two-minute trial so you know the room value, print the sheets.

Sensor settling time — **60–90 seconds** after every move, up to three minutes if the temperature difference is large. It is the slowest sensor in the Junior activities and the single most likely reason for bad data.

Sensors and boxes — BME280. One box is enough for the whole class if you work in a circle with the screen projected; ideally, one box per 4–6 children.

Three routes

Short	20 min: written prediction, room temperature, temperature beside the heat source, comparison and conclusion. **Exactly two places.**
Full	The 50-minute lesson: the short route, unhurried, plus two more places from the room map and the discussion of humidity.
Project	The full lesson, then the vertical map (lesson 2) and recording a whole lesson (lesson 3).

1. Where this fits in the curriculum

Every child already knows what "hot" and "cold" mean. What they do not know is that the room they are sitting in has, at the very same moment, several different temperatures, and that the difference can be measured. That discovery — that a room is not one uniform thing — is the heart of this activity and the beginning of real thermal reasoning. The activity fits:

- **Mathematics and exploring the environment** (*Matematică și explorarea mediului*, Romanian school years I–II) — this is where the year II part sits naturally: objects and phenomena from the immediate environment, observing the weather, measuring with an instrument and comparing two values. At this level the child records only the whole-number part of the temperature.
- **Natural sciences** (*Științe ale naturii*, years III–IV) — temperature as a measurable property, heat sources, warm air rising and cold air sinking, evaporation and condensation as a first encounter with humidity, and the idea of a *fair comparison*: same sensor, same waiting time, different place.
- **Mathematics** (*Matematică*, a subject in its own right from year III) — reading a scale, subtracting to find a difference, ordering values. **Decimal fractions do not enter the syllabus until year IV**, so in years II–III the temperature is recorded rounded to whole degrees; in year IV, 21.4 °C becomes a real example of a decimal fraction, not an exercise from the textbook.
- **Geography** (*Geografie*, year IV only, where the subject appears in the Romanian national timetable) — the weather and how it is recorded, why weather stations exist, why a thermometer must be kept in the shade to be honest.
- **Visual arts and practical skills** — drawing the warm place and the cool place, and the thermal map of the room. This is an *application*, not a syllabus entry: the drawing is here the means by which the child organises the data, not the art lesson's own subject.
- **Cross-curricular** — the patience to wait for an instrument, recording what the screen shows rather than what you expected, and a first idea about a thermostat, a fridge and a radiator.

The core skill is **controlled comparison**: the same instrument, treated the same way, moved to a different place. Children of this age can grasp completely that a comparison is only fair if a single thing changes.

2. Time plan

The in-app activity has six steps: the introduction, the prediction, two measurements, a short "How sure are we?" step and the conclusion. It fits into a quarter of an hour *only* if the sensor's waiting time is made visible and the number of places measured is capped at two. That is why the short route below is twenty minutes, not fifteen: the minute of settling the app asks for at every measuring step is real and must be planned for, not skipped.

Route	Time	What happens
Short	**20 min**	Written predictions (3), the introduction on screen (2), room temperature with the waiting made visible (6), temperature beside the heat source (6), "How sure are we?", comparison and conclusion (3). These are exactly the app's steps, with the prediction it asks for before measuring, and **exactly two places measured**.
Full	**50 min**	Written predictions and their reasons (8), the introduction and the temperature reference points (4), room temperature (7), beside the heat source (8), two more places chosen by the class (12), comparison, drawing and conclusion (11).
Project, lesson 2	50 min	The vertical map: floor, desk height, on top of the cupboard. Three heights, with two readings at each: about eight minutes per height, with the move, the wait and the two readings, so $3 \times 8 = 24$ minutes; the class's shared graph on the board (13), the written conclusion (13).
Project, lesson 3	50 min	Change over time: one box logs for the whole lesson while the class works on something else, then the graph is read together and the class looks for the moment the door opened or the sun came out. Optionally, the damp cloth and humidity.

The project route begins with the full lesson and adds lessons 2 and 3, so it occupies **three 50-minute lessons in the same week** — two and a half hours of contact time. No homework and no preparation at home is asked of the children.

The visible timer and a task for the waiting time

The temperature sensor is **slow** — the biggest practical difference from the light activity. The BME280 sits inside a plastic housing that must itself warm up or cool down, and that takes between one and three minutes. A child who moves the box and reads the number after five seconds records a wrong value and concludes that nothing changed.

The remedy has two halves, and both are compulsory.

First, make the waiting visible. The app has its own timer, above the chart: **Let the box settle: … s more** counts 75 seconds and turns green once the reading has settled. After every move press **I moved the box** to start it again. Project the screen so the whole class sees it; without a projector, a large kitchen clock set to **60–90 seconds** does the same job. Children have no patience for an unseen wait, but they do have patience for one that visibly counts down.

Then, fill the wait with a task. During the 60–90 seconds, every child does *one* of the following things, announced beforehand: writes the prediction for the **next place**, draws on the sheet where in the room the just-measured spot is, or reads the number on screen aloud once every fifteen seconds, so the whole class hears it approaching its final value. Watching a number crawl towards its true value is, by itself, a very good thing for a nine-year-old to see.

3. Materials

What you need	Per group	Notes
MagicBox with BME280	1	Gives temperature, humidity and pressure. The activity uses the first two.
Laptop with the app	1	By USB cable or over WiFi. A long cable, or WiFi, is what lets you reach the radiator and the window.
A timer the whole class can see	1	The app's own timer, projected, or a kitchen clock. It is not an accessory: without it the 60–90 second wait does not get respected.
A heat source that actually works	1	The class laptop, a radiator in winter, or a desk lamp switched on five minutes beforehand. The app proposes the computer first, since every classroom has one. **Check it with your hand before the lesson:** a cold radiator in June gives no result.
Jug of warm tap water	1	**Warm, not hot, and never water from a kettle.** Around 40 °C is plenty and safe to touch. For anything warmer, the **teacher** demonstrates **at the front, with the children seated**.
Bowl of ice cubes	1	For the cold end of the scale. The sensor sits *above* the ice, never in the water.
Cloth or paper towel, dampened	1	Placed near the sensor, not on it, it raises the humidity. A striking demonstration of evaporation.
Dry towel	1	For the insulation extension and for drying hands and surfaces.
Sticky labels	4–6	To mark and name the measuring spots, so that every group measures the same places.
Ordinary classroom thermometer	1	Optional but valuable: comparing the sensor with a familiar thermometer builds trust in the instrument.
Tray for liquids	1	The warm water and the ice sit on the tray, on one side of the table; the box and the laptop sit on the other.
Clipboard, coloured pencils	1 / child	The sheet is filled in standing up, and the drawing calls for blue and red.

Preparation the day before: switch the heating on or off deliberately so that you know what the room will do, and decide in advance which heat source you will use. Stick the labels on the measuring spots and write the name of the place on each, so that every group measures the same points. Leave the boxes in the classroom overnight, not in a cold store cupboard, so that they start at room temperature. If you are using ice, freeze it the day before and have a tray and a cloth ready for spills. Finally, leave a box on the desk for two minutes and note the room value: it lets you tell during the lesson whether a group's number is plausible.

4. Scientific background for the teacher

Temperature, heat, and why they are not the same thing

Temperature measures the average kinetic energy of the particles in a substance. **Heat** is energy in transit, from a warmer body to a cooler one. A bathful of water at 30 °C contains far more energy than a cup at 60 °C, and yet the cup is hotter. Children conflate the two constantly, and the teacher needs to be clear even though the word "heat" will be used loosely in class.

The sensor does **not** measure the air's temperature directly. It measures *its own* temperature and reports that. The value shown is the air temperature only once the sensor has reached thermal equilibrium with the surrounding air — the zeroth law of thermodynamics in action, and the physical reason the app asks for a minute of waiting. The approach to equilibrium is exponential:

$$T(t) = T_{\text{air}} + \left(T_0 - T_{\text{air}}\right) \mathrm{e}^{-t/\tau},$$

where τ is a time constant set by the sensor's thermal mass and how well air moves past it. For a small board in a plastic case, in still air, τ is of the order of 30–60 seconds. After 60 seconds the reading has covered about three quarters of the gap, after 90 seconds nearly nine tenths, and after three minutes essentially all of it. This is why a hurried measurement comes out too low when you move the sensor somewhere warm, and too high when you move it somewhere cold.

4.1 Why the classroom does not have one temperature

Three mechanisms move heat, and all three are visible in a classroom.

Conduction carries heat through solid material: the metal leg of a chair feels colder than the wooden seat at the same temperature, because it takes heat from your hand faster. **Convection** is the main mechanism here. Air warmed by a radiator or a computer expands, becomes less dense and rises; cooler air flows in underneath to take its place. The result is a permanent slow circulation and a vertical difference of typically 1 to 3 °C between floor level and head height in a heated room — considerably more above a radiator. **Radiation** needs no air at all: any warm object emits infrared, which is why you feel the sun through a window and why a sensor left in direct sunlight reads far above the surrounding air. The Stefan–Boltzmann law gives the power emitted per unit area:

$$P = \varepsilon \sigma T^4,$$

with $\sigma = 5.67 \times 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}$. The fourth power is why an object only a little hotter radiates so much more.

A laptop turns almost all the electrical power it draws into heat — for an ordinary school machine, of the order of 20 to 50 watts under load. That is comparable to a small light bulb left running, and it is entirely enough to raise the air a few centimetres above the keyboard by one to three degrees. The children's discovery at the fourth step of the app is, therefore, genuine physics, not an artefact.

The arithmetic for the whole class is worth doing too: a person at rest **gives off about** 60–80 **watts** of metabolic heat — an adult nearer 80, a child nearer 60 —, partly by radiation, partly by warming the air around them and partly through breathing and sweat. Thirty children therefore give about $30 \times 65 \approx 2000$ watts, that is **about two kilowatts** — as much as an electric heater, switched on for the whole lesson. The room warms up visibly over the course of the lesson, and the children are the cause.

4.2 Relative humidity and why it moves when temperature moves

The BME280 reports **relative humidity**: the amount of water vapour in the air, expressed as a percentage of the maximum the air could hold at that temperature.

$$\mathrm{RH} = \frac{e}{e_s(T)} \times 100\%,$$

where e is the actual vapour pressure and $e_s(T)$ the saturation vapour pressure. The point that matters for teaching is that e_s rises steeply with temperature — roughly doubling for every 10 °C. Consequently, if you warm a parcel of air without adding any water to it at all, the relative humidity *falls*. Children will see this the moment the sensor goes near the computer: the temperature climbs, the humidity drops, and they will ask why. The answer is that the air has not lost a single drop of water; it has simply gained the capacity to hold more. This is also why heated buildings feel dry in winter, and why a cold window pane runs with condensation — the air touching it has been cooled below its dew point.

The comfort band quoted in the app, 40–60%, is the range in which respiratory irritation, static electricity, mould and dust mites are all reasonably low. It is a genuine public-health figure, not a rule of thumb.

4.3 Fixed points worth knowing

The Celsius scale is anchored on water: 0 °C the freezing point and 100 °C the boiling point, at standard atmospheric pressure. The app lists 37 °C for the human body and 20–22 °C for a comfortable room. Absolute zero, at −273.15 °C, is where the Kelvin scale begins; $T_{\mathrm{K}} = T_{\mathrm{C}} + 273.15$. The fact that there is a coldest possible temperature but no hottest one delights this age group and costs thirty seconds.

5. Lesson flow

The in-app wizard has six steps: an introduction, a **prediction**, room temperature, a measurement beside a heat source, the **How sure are we?** step and a conclusion by comparison. At the prediction the children write (or dictate) their answer to the on-screen question in a text box; the app keeps it in the journal and quotes it in the conclusion. Below is what the teacher does around them. The times are for the 20-minute short route; on the full 50-minute lesson each step is given room to breathe and two more places are added.

1. **Ask for the written predictions first (3 min).** Before any box is switched on, ask: *is the temperature the same everywhere in this room?* Take a show of hands and write the split on the board. Then every child writes on the sheet how warm they think the room is, and whether the air beside the computer will be warmer, cooler or the same — and by how much. Most will say “the same everywhere”. That written commitment is what makes the result matter fifteen minutes later.
2. **Introduction on screen (2 min).** Read the four reference points together: freezing at 0 °C, a comfortable room at 20–22 °C, body temperature at 37 °C, boiling water at 100 °C. Ask where they think today’s room will land, and have them mark the spot on the sheet’s scale. The activity is opened from the **Activities** page, with the **Start** button on the **The Magic Thermometer** card.

Technical note — right before the first reading

The sensor. BME280, which gives temperature, relative humidity and pressure from a single chip. Nothing extra is plugged in. The activity uses the first two quantities; the cards on screen are **Temperature** and **Humidity**, with their chart underneath, covering the last three minutes. The activity is a Junior one, so the temperature is shown with one decimal,

the humidity as a whole number, and there are no smoothing, pause or clear buttons above the chart.

How long to wait — the one thing that ruins this activity's data. 60–90 seconds after every move, up to three minutes if the temperature difference is large. The sensor does not measure the air, it measures itself, and its plastic housing must first reach the air's temperature. The app counts 75 seconds by itself (**Let the box settle**); after every move press **I moved the box** to restart it, and give the children a task for the duration — see the teaching tip in the time plan. When it turns green (**the reading has settled**), press **Capture** twice, about 30 seconds apart: the app uses the mean of the two and compares them at the **How sure are we?** step.

The buttons you will use. The activity moves forward with the green button at the bottom right, and **its label changes at every step**, because it echoes what you have just finished: **Let's start!**, **My prediction is written**, **Room temperature noted**, **Temperature near computer noted**, **Continue**, and at the last step **Finish**. The **Back** button, on the left, goes back one step. If you move on with no value captured, the app reminds you to press **Capture** first.

Exporting the data. Two buttons appear on the measurement steps: **PNG** saves the chart as an image, and **Excel** saves the step's raw readings to an `.xlsx` file. At the end the app saves the activity's journal (the prediction, the notes and the captured values) in the `logs` folder by itself. **There is no CSV button inside the activity.** The whole recording is taken from the **Data & Export** page (shown at the Scientist and Teacher levels, chosen in **Settings**), with the **Excel** button on the box's row; recordings are never deleted automatically. Save the file using the pattern `class_date_activity` — for example `3B_2026-09-14_termometrul_magic`. Note on the sheet the place measured (the name on the label and its height above the floor), the time, whether the heating was on and whether the window was open, and check that the file really was saved before closing the application.

3. **Room temperature (6 min).** The box is placed on a table, away from windows, radiators, laptops and children's hands, and stays there for **60–90 seconds**, with the timer running and visible. Use the wait: one child reads the value aloud once every fifteen seconds, so the whole class hears it settling, while the rest write the prediction for the next place. Then press **Capture** twice and record both the temperature and the humidity. Ask whether today's humidity is inside the comfortable 40–60% band.

4. **Beside the heat source (6 min).** Move the box next to the computer's air vent or to the radiator and press **I moved the box**. Watch the chart climb — this is the visual pay-off of the activity and worth a pause to look at. Wait 60–90 seconds again, capture the value twice, then record the new temperature and humidity. Ask the app's own question: *why is it warmer here?* The machine uses electricity and gives off heat. Then ask the harder, better question: *what happened to the humidity number, and why?*

5. **Comparison and closing (3 min).** The **How sure are we?** step shows the two readings captured at each place and says whether the difference between the places is bigger than the gap between two readings of the same place (so it is real) or whether the class should measure again. Read it aloud: it is the children's first meeting with the idea of uncertainty. Then subtract the two temperatures on the board to get the difference; the app's conclusion shows it too, next to the children's prediction. Fix the conclusion in the children's own words: a room does not have one temperature, and anything that uses energy warms the air around it. Close with the app's conclusion step, whose green button reads **Finish**, and its outward links — the thermostat at home, the weather station, the fridge. Finish by asking every child to complete the conclusion sentence on the sheet.

If you have twenty extra minutes

Three extensions, in rising order of interest. First and best is the **vertical map**: floor, desk height, on top of the cupboard — usually between 1.5 and 5 °C between the lowest and the highest, and children genuinely find it astonishing that the air near their feet is colder than the air near their head. Second, the **damp cloth**: placed near the sensor, it raises the humidity within a few minutes as the water evaporates, and the temperature may even dip slightly, because evaporation uses energy. Third, the **insulation test**: wrap the box loosely in a dry towel, next to the radiator, and compare how fast it warms against an unwrapped box. This is the physics of a coat, a duvet and a house.

The investigation, in one table

The question. Does our room have the same temperature everywhere?
The hypothesis. Written on the sheet **before** any measurement, with a number for the room temperature and a "warmer / cooler / the same" for the place beside the heat source, plus a reason in a few words. It is not changed afterwards.
What I change. A single thing: the **place** where the box sits.
What I measure. Temperature, in °C, and humidity, in percent, exactly as they appear on the cards on screen.
What I keep the same. The same box, the same waiting time at every place, the same person reading the screen, the box placed on a surface rather than held in the hand, and nobody breathing or leaning over it.
How many repeats. Two readings for each place, thirty seconds apart, *after* the waiting time is over. If the two are close, the value is good; if they differ a lot, the sensor had not settled yet.
What makes a reading valid — and what makes it an artefact. A valid reading is one taken after 60–90 seconds, with the box on a surface and nobody above it. Anything else is an artefact: a box held in the palm gains a degree or two within a minute, a breath over it spikes both the temperature and the humidity at once, and a box moved and read after ten seconds still shows the temperature of the previous place. The class rule: move it, start the timer, wait, only then read. **If two groups get very different values for the same labelled place, the cause is almost certainly the waiting time, not the sensor** — repeat the measurement together, with a single timer for the whole class.

6. Analysis after the activity

6.1 What the data should look like

Place	Typical value	What it means
Open table, away from sources	19–23 °C, 35–55% RH	The room's baseline; the number a thermostat would act on.
At floor level	0.5–2 °C below the table	Cool air sinks.
On top of a cupboard	1–3 °C above the table	Warm air rises.
Beside a laptop's air vent	2–6 °C above the table	Electrical energy turned into heat.
Directly above a hot radiator	5–15 °C above the table	Strong convection current.
Humidity beside the radiator	3–10 percentage points lower	Warmer air can hold more water, so relative humidity falls.
Above a bowl of ice, not touching	3–10 °C below the table	Slow; allow several minutes.
Near a damp cloth	humidity up by 5–20 points	Evaporation.

The two rows of the vertical map read together: the floor sits 0.5–2 °C below the table, and the top of the cupboard is 1–3 °C above it, so **between the floor and the cupboard you get between** 1.5 **and** 5 °C. This is the figure children will find the most surprising in the whole activity.

Absolute values depend on the season, the heating, the weather and how many children are in the room. Do not chase a "correct" temperature. What must come out reliably is the **pattern**: warmer near sources, warmer higher up, and humidity moving in the opposite direction to temperature.

6.2 Questions to ask afterwards

1. Before we measured, most of us thought the whole room was the same temperature. What did the sensor show, and where was the biggest difference?
2. When we moved the sensor to the warm place, the number did not jump straight away — it climbed slowly. Why do you think it took time?
3. The temperature went up beside the computer, and the humidity went down. Did the air lose water? What else could explain the drop?
4. Where in this room would you put a plant that likes warmth? Where would you put a snowman? Use your numbers to justify both answers.
5. If we measured again tomorrow morning, before anyone arrived, which of our numbers would be different, and which would be about the same?

6.3 Ideas for assessment along the way

- **Low stakes:** the child circles "cool", "warm" or "hot" on the sheet for each place, records the whole-number part of the temperature, and draws the two places, colouring the warm one red and the cool one blue. Decimals are not required.
- **Standard:** the child records both temperatures and both humidity values, works out the difference by subtraction, and writes one full sentence naming the heat source and saying why it warms the air.

- **Extension:** the child predicts a value for a third place before measuring it, checks it, explains any mismatch, and builds the **vertical map** of the room — floor, desk, cupboard — then writes why the humidity fell as the temperature rose, using the idea that warm air can hold more water.

Common pitfalls and misconceptions

Reading too soon. By far the most common mistake. The sensor needs 60–90 seconds in a new place, sometimes three minutes. A child who reads after ten seconds records a value halfway up the climb, and two groups doing this get completely different answers for the same place. Enforce the wait and make it visible.

Holding the box in warm hands. A hand around the case adds a degree or two within a minute. So does breathing on it, which spikes both the temperature and the humidity at once. The box goes on a surface and stays there.

"Warm and hot are the same word." Children will call a 24 °C reading "hot". Anchor the vocabulary to the app's reference points: hot is nearer 37 °C, boiling is 100 °C, and 24 °C is a pleasant room.

Confusing temperature with how something feels. Metal at 21 °C feels colder than wood at 21 °C, because it takes heat from the skin faster. Children will insist the metal table leg is colder; the sensor placed on each in turn is the best possible answer — but remember it must be left to settle on each.

Confusing heat with light. If you have also run the light activity, children may assume "bright" means "warm". A cool LED lamp and a warm, dark radiator settle it in ten seconds.

Expecting humidity to rise with temperature. It is intuitive, and it is wrong. Relative humidity usually *falls* when the air is warmed — with one exception: beside warm water, where water evaporates into the air, humidity rises. Have the explanation ready before the lesson, because a sharp child will ask.

Sunlight on the sensor. A box left in a patch of direct sun reads far above the true air temperature, because it is being heated by radiation. Real weather stations shield their thermometers for exactly this reason — a good thing to tell the class, not to hide.

Too many places in a short lesson. The twenty-minute route has room for **two** measurements, not five. A class that tries five places in twenty minutes waits at none of them, and goes home with five wrong numbers instead of two good ones.

Safety and classroom management

Warm water is the main hazard. Use warm tap water, around 40 °C, **never water from a kettle**. The teacher pours; children do not carry open containers of warm water around the room. If the demonstration needs anything hotter, the **teacher** does it **at the front, with the children seated**, and nobody else touches the container.

Radiators can be genuinely hot in a well-heated building. Check the surface with your own hand before letting a child place the box near it, and if it is too hot to touch comfortably, measure the air a hand's width above it rather than on it. No child leans or climbs on a radiator.

Electronics and water do not mix. Ice, damp cloths and jugs make puddles. Keep all liquids on a tray, keep the box and the laptop on a different part of the table from the water, and dry hands before touching the equipment. **The box is not waterproof**, and the sensor is never dipped into anything.

Ice straight from the freezer can stick to skin. Hand out cubes that have sat outside for a minute, and let children touch them briefly or with a spoon.

Cables. A USB cable stretched from the laptop to the radiator or the window is both a trip hazard and a way of pulling the laptop off the table. Use WiFi where you can, or tape the cable down and keep the walking route clear.

Moving around the room. The activity sends children to the edges of the room. Set a walking rule and a maximum of two children per box before anyone stands up.

Adaptations for pupils with special educational needs

No decimals at all. This is the main adaptation, and it is not a concession: in years II–III decimal fractions are not even in the syllabus. The child records only the whole-number part (writes 21 when the screen shows 21.4), or not even that — circling "cool", "warm" or "hot" on the sheet. The goal, that different places have different temperatures, is met without a single decimal. The sheet has its boxes already printed for exactly this reason, and the child can colour the cool place blue and the warm place red instead of writing anything.

One box, the whole class. Sit the children in a semicircle, with the screen projected, and move the box yourself from place to place. Every move has a named **screen reader**, who says the number aloud, and a **timer keeper**, who announces when the 60–90 seconds are up. Roles rotate at every place, so every child has a real task, not a wait.

Three real roles for children with motor difficulties. Not being able to carry the box does not mean not taking part. Roles rotate at every place:

- **timer keeper** — starts the timer, announces aloud when the wait is over, and has the right to stop any reading taken too soon; this is the role with the greatest scientific value in the group, because it decides when a reading counts;
- **screen reader** — says the two numbers aloud, once every fifteen seconds during the wait and once more at the end;
- **mapper** — draws on the sheet where in the room the measured place is, and colours it blue or red.

The box can be carried by a classmate, on the child's instruction: "put it closer to the radiator" is an experimental decision too, and whoever makes it is running the experiment.

Support with writing. Spoken answers are accepted instead of written ones, with the teacher or a classmate writing them on the sheet. For the prediction, the child may tick one of the boxes already printed instead of writing a free-form answer. The sheet comes with its labels, rows and drawing box already printed for exactly this reason.

Language, sensory needs and patience. Use the same three words all lesson (temperature, humidity, sensor) and always pair them with the same gesture. For children who find waiting hard to bear, give them the role of timer keeper: the wait becomes their task, not their ordeal. Announce in advance that the number on the screen **will move** for a whole minute and that this is normal — an anxious child watching an unstable number may otherwise believe they have broken the instrument.

If things do not go to plan

You have one box and thirty children. Semicircle, screen projected, and you move the box yourself. Two places, two timed waits, one named screen reader and one timer keeper at each place. The table is filled in together, on the board, and the children copy it onto their sheets.

You have twenty minutes, not fifty. This is exactly the short route, and it works, on one condition: **exactly two places** — the open table and the heat source. Drop the room map, the ice and the damp cloth. The written prediction at the start and the calculated difference at the end are all the lesson needs for a result. The conclusion is said aloud, not written.

You have no projector. Sit the class in a semicircle around the laptop, close enough to see the screen, and name a **screen reader** at each place, who says the numbers aloud for everybody. It works just as well, and sometimes better: somebody has to pronounce every value.

The heat source does not warm anything. A cold radiator in June or an LED lamp gives no result. The safe fallback is a jug of warm tap water, around 40 °C, placed on a tray, with the sensor *beside* it, not in it. Note that beside warm water the humidity *rises*, because water evaporates — unlike beside the computer or the radiator. If even that is not possible, use the class laptop: after ten minutes of use its air vent is a perfectly honest heat source.
The sensor will not stop climbing or falling. This is normal in the first minute. If the number is still drifting after three minutes, look for a draught: a window left ajar or a door that keeps opening stops the sensor from reaching equilibrium. Close them and start again.
The box will not connect. Do not stop the lesson. Use the "What the data should look like" table from this guide as a printed data set: the children write their predictions first, then receive the values row by row and check whether they guessed the warmer place. The discussion of why a room does not have one temperature carries on unchanged. Repeat the measurement on another day, after checking the cable, the port selected in the app and, if you are on WiFi, the network.

7. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are root domains, not deep links, so they will not rot over time.

- **National Meteorological Administration (ANM)** — `https://www.meteoromania.ro` — search for today's weather and the weather stations; children can compare the room's temperature with the one outside, measured by a real station, at the same hour.
- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for the energy forms and energy transfer simulation, which animates exactly the conduction and convection children have just measured and lets them see the particles moving.
- **NOAA** — `https://www.noaa.gov` — search for how real weather stations measure temperature and humidity, and why a thermometer is kept in a shaded, ventilated screen. Fits perfectly alongside the sunlight-on-the-sensor pitfall.
- **Institute of Physics** — `https://www.iop.org` — search their primary-level material on heat and temperature; it is written for non-specialist teachers and addresses the heat-versus-temperature confusion directly.
- **STEM Learning** — `https://www.stem.org.uk` — search for heating and cooling, or insulation, for ages 7–11; a large library of lesson plans this measurement can be dropped straight into.
- **World Health Organization** — `https://www.who.int` — search for indoor environment and housing, for the public-health angle on comfortable indoor temperature and humidity.

8. Assessment

The exit ticket, identical to the one on the student sheet: *one thing that surprised you today.* It is collected in the last two minutes of the lesson and says more about the class's understanding than any mark.

For a more formal judgement, use the **shared three-strand rubric** — *measurement practice*, *data interpretation* and *scientific reasoning.* At Junior level, the three strands look like this:

- **Measurement practice** — the child waits for the timer to finish before reading, leaves the box on a surface, and records what the screen shows rather than what was expected.
- **Data interpretation** — the child says which place was warmer and by how much, and notices that the humidity moved the opposite way.

- **Scientific reasoning** — the child explains, in their own words, why a room does not have one temperature, and compares the conclusion with the prediction written at the start.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. The student sheet

The sheet the children fill in is a separate document: `docs/en/student_sheets/magic_thermometer.pdf`. It has two pages — the prediction, the scale of reference points and the map box on the first; the measurement and the conclusion on the second — so it prints double-sided, one sheet of paper per child. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write — and asks for the prediction **before** any measurement.

Three things worth knowing before printing it. **The drawing box comes before the table, not after it:** the room map is the task that fills the 60–90 seconds of waiting, and the child writes the temperatures beside the points already drawn. **The table has four rows already labelled** — table, heat source, floor, on top of the cupboard — and the last two are exactly the vertical map; the short route fills in only the first two. **The "cool / warm / hot" column** is there precisely for the child who does not yet work with decimals: they circle it and leave the numbers column blank. The table is not repeated here precisely so that the two cannot drift apart: there is a single printable version, and it is the sheet.